

Question 1

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

- A. find the disruptions more irritating as time went on.
- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
- C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.
- D. perceive the volume of the construction noise as growing softer over time.

Question 2

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht's influence can be seen in English playwright Caryl Churchill's 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

- A. focus on the characters' beliefs about social and political issues as revealed by the characters' actions.
- B. reflect on social and political phenomena not directly related to patriarchy and colonialism.
- C. recognize pertinent social and political parallels between Germany during Brecht's time and England at the time when Churchill was writing *Cloud 9*.
- D. be dispassionate as they think critically about the social and political questions raised by the play.

Question 3

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn't recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn't already seen it, the whalelike geoglyph _____

Which choice most logically completes the text?

- A. must represent a species of whale that went extinct before there were any people in Peru.
- B. is actually located in Germany, not Peru, and isn't part of the Nazca Lines at all.
- C. is probably in a location Isla hadn't ever come across while on the ground.
- D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

Question 4

Indigenous cultures possess unique knowledge of the medicinal uses of plants. According to a 2021 study, 73 percent of the medicinal uses of plants native to North America are reflected in the vocabulary of a single Indigenous language. However, as more and more Indigenous people exclusively speak a globally dominant language, such as English, their ancestral languages fade from daily use. These facts lend added importance to tribal nations' efforts to preserve their languages. By ensuring the continued use of Cherokee, Ojibwe, and the hundreds of other Indigenous languages in what is now the United States, tribal nations are also _____

Which choice most logically completes the text?

- A. increasing the number of medicinal plants represented in the vocabularies of Indigenous languages.
- B. transmitting terms for medicinal plants from Indigenous languages to globally dominant languages.
- C. preserving knowledge about the medicinal value of plants native to the tribal nations' lands.
- D. ensuring that citizens of tribal nations have physical access to medicinal plants.

Question 5

Henry Ossawa Tanner's 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

- A. risk judging Tanner's painting by standards that may not be historically appropriate.
- B. tend to conflate Tanner's political views with those of his contemporaries.
- C. forgo analyzing Tanner's painting in favor of analyzing his political activity.
- D. wrongly assume that Tanner's painting was intended as a critique of his fellow artists.

Question 6

A heliograph is a semaphore device used for sending optical communications—usually in the form of Morse code—by reflecting flashes of sunlight off a mirror. Heliographs were used for rapid communication across expansive distances for military, surveying, and forestry purposes during the late nineteenth and early twentieth centuries, but they were largely effective only during the daytime, and the range of the device depended on factors such as the opacity of the air and line of sight. Therefore, heliographs were eventually replaced by technology that _____

Which choice most logically completes the text?

- A. worked on similar principles but was easier to produce and maintain.
- B. was not so constrained by environmental circumstances.
- C. could be used for more than military, surveying, or forestry purposes.
- D. enabled communication that didn't require knowledge of Morse code.

Question 7

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because _____

Which choice most logically completes the text?

- A. it can only be used on burrows below a certain size.
- B. it does not allow for multiple castings of the same burrow over time.
- C. the casting process takes more time than *A. bellulus* takes to construct a burrow.
- D. the process of recovering the model distorts the resin's shape.

Question 8

Conventional theories of rhetoric hold that presenting information as coming from credentialed experts increases that information's credibility. When communications researcher Sungkyoung Lee and her colleagues tested messages seeking volunteers for clinical trials, however, they found that participants in their study judged recruitment messages from former trial volunteers as significantly more credible than messages from doctors (i.e., credentialed experts). One reason for this may be that the doctors' status as credentialed experts wasn't ignored but rather was outweighed by participants' views of the experiential relevance of the two types of messengers; that is, participants may have reacted the way they did because _____

Which choice most logically completes the text?

- messages from former trial volunteers depicted clinical trials as being more positive experiences than did messages from doctors.
- participants did not have enough experience to evaluate the credibility of the doctors' messages but did have enough experience to evaluate the credibility of former trial volunteers' messages.
- the fact that former trial volunteers went through the same experience that participants were contemplating while doctors did not was more important to participants than the doctors' status as credentialed experts was.
- participants regarded the experiences of both the doctors and former trial volunteers as relevant to the subject of clinical trials but were skeptical of the doctors' status as credentialed experts.

Question 9

During their larval phase, numerous species of coral reef fish are drawn toward areas where light is present. To better understand how artificial light at night (ALAN) might affect some coral reef fish, researchers explored the effect of exposure to low levels of ALAN on the reproductive success of the common clownfish (*Amphiprion ocellaris*). While exposure to low levels of ALAN had no significant effect on spawning frequency and egg fertilization in *A. ocellaris*, incubation in the presence of ALAN completely inhibited hatching. These findings suggest that _____

Which choice most logically completes the text?

- A. ocellaris* that settle in areas with low levels of ALAN have significantly higher rates of successful egg fertilization than *A. ocellaris* that settle in areas without ALAN do.
- the reproductive success of *A. ocellaris* would be at risk if they were to selectively settle in regions that are regularly exposed to low levels of ALAN.
- the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN during incubation than the reproductive success of other species of coral reef fish is.
- the spawning frequency of *A. ocellaris* was more strongly affected by the presence of low levels of ALAN than egg fertilization was, though both were less affected than incubation.

Question 10

Overgrazing by purple sea urchins has caused many kelp forests along North America's west coast to be replaced by urchin barrens—areas stripped of vegetation and covered in purple sea urchins. Urchins in barrens persist in a state of starvation that lessens their nutritional value—and thus their appeal—to many predators. Sarah Gravem and colleagues placed sunflower sea stars, a once-abundant predator species suffering massive population declines in recent years, in aquariums that each contained a nutritionally poor and a nutritionally rich purple sea urchin. The researchers found that the sea stars selected the nutritionally rich urchin in 42.7% of trials and the nutritionally poor urchin in 37.5% of trials, suggesting that _____

Which choice most logically completes the text?

- sunflower sea stars are willing to hunt sea urchins, but if given a choice, they will prey on other more nutritious marine animals instead.
- sunflower sea stars are reluctant to feed on both nutritionally poor and nutritionally rich sea urchins and are therefore unlikely to thrive in kelp forests.
- sunflower sea stars are less likely to consume sea urchins in barrens than other species of sea stars are, putting sunflower sea stars at a high risk of extinction.
- sunflower sea stars do not always avoid foraging on nutritionally poor sea urchins, making sunflower sea star population recovery a potentially important tool for controlling urchin barrens.

Question 11

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylander* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host's fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylander* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the _____

Which choice most logically completes the text?

- uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected workers in the nest.
- need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.
- high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly disperse when the nest is attacked by a predator.
- average lifespan of *T. nylander* worker ants in colonies without parasitic activity typically falls well below three years, the range covered by the study.

Question 12

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

- those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.
- the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.
- Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.
- Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.

Question 13

One theory behind human bipedalism speculates that it originated in a mostly ground-based ancestor that practiced four-legged "knuckle-walking," like chimpanzees and gorillas do today, and eventually evolved into moving upright on two legs. But recently, researchers observed orangutans, another relative of humans, standing on two legs on tree branches and using their arms for balance while they reached for fruits. These observations may suggest that _____

Which choice most logically completes the text?

- A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.
- B. bipedalism must have evolved simultaneously with knuckle-walking and tree-climbing.
- C. moving between the ground and the trees would have been difficult without bipedalism.
- D. a knuckle-walking human ancestor could have easily moved bipedally in trees.

Question 14

The increased integration of digital technologies throughout the process of book creation in the late 20th and early 21st centuries lowered the costs of book production, but those decreased costs have been most significant in the manufacturing and distribution process, which occurs after the authoring, editing, and design of the book are complete. This suggests that in the late 20th and early 21st centuries, _____

Which choice most logically completes the text?

- A. digital technologies made it easier than it had been previously for authors to write very long works and get them published.
- B. customers generally expected the cost of books to decline relative to the cost of other consumer goods.
- C. publishers increased the variety of their offerings by printing more unique titles but also printed fewer copies of each title.
- D. the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.

Question 15

Volunteering, or giving time for a community service for free, is a valuable form of civic engagement because helping in a community is also good for society as a whole. In a survey of youths in the United States, most young people said that they believe volunteering is a way to help people on an individual level. Meanwhile, only 6% of the youths said that they think volunteering is a way to help fix problems in society overall. These replies suggest that _____

Which choice most logically completes the text?

- A. many young people think they can volunteer only within their own communities.
- B. volunteering may be even more helpful than many young people think it is.
- C. volunteering can help society overall more than it can help individual people.
- D. many young people may not know how to find ways to volunteer their time.

Question 16

Ancestral Puebloans, the civilization from which present-day Pueblo tribes descended, emerged as early as 1500 B.C.E. in an area of what is now the southwestern United States and dispersed suddenly in the late 1200s C.E., abandoning established villages with systems for farming crops and turkeys. Recent analysis comparing turkey remains at Mesa Verde, one such village in southern Colorado, to samples from modern turkey populations in the Rio Grande Valley of north central New Mexico determined that the latter birds descended in part from turkeys cultivated at Mesa Verde, with shared genetic markers appearing only after 1280. Thus, researchers concluded that _____

Which choice most logically completes the text?

- A. conditions of the terrains in the Rio Grande Valley and Mesa Verde had greater similarities in the past than they do today.
- B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.
- C. Indigenous peoples living in the Rio Grande Valley primarily planted crops and did not cultivate turkeys before 1280.
- D. the Ancestral Puebloans of Mesa Verde likely adopted the farming practices of Indigenous peoples living in other regions.

Question 17

Many of William Shakespeare's tragedies address broad themes that still appeal to today's audiences. For instance, *Romeo and Juliet*, which is set in the Italy of Shakespeare's time, tackles the themes of parents versus children and love versus hate, and the play continues to be read and produced widely around the world. But understanding Shakespeare's so-called history plays can require a knowledge of several centuries of English history. Consequently, _____

Which choice most logically completes the text?

- A. many theatergoers and readers today are likely to find Shakespeare's history plays less engaging than the tragedies.
- B. some of Shakespeare's tragedies are more relevant to today's audiences than twentieth-century plays.
- C. *Romeo and Juliet* is the most thematically accessible of all Shakespeare's tragedies.
- D. experts in English history tend to prefer Shakespeare's history plays to his other works.

Question 18

In a 2017 article, historian Jasmine Kilburn-Toppin explains that in early modern London, members of the city's guilds (trade and artisanal associations) were participants in a civic culture in which gift giving both signaled and conferred social status. Research on this phenomenon has tended to focus on philanthropic gifting by London's largest guilds; for her part, Kilburn-Toppin focuses on the gifting of handmade objects and fixtures (such as decorative paneling or plasterwork) within the craft guilds, which were "composed of highly discerning producers and consumers of material cultures." Given this characterization, it can reasonably be inferred that the gifting of such objects may have _____

Which choice most logically completes the text?

- A. ensured that knowledge of the materials and techniques used by members of the craft guilds stayed within the craft guilds.
- B. emphasized ostentatious offerings in the case of the largest guilds and prosaic objects in the case of the craft guilds.
- C. functioned as a way for craft guild members to maintain and enhance their professional reputations among their peers.
- D. conferred greater social status when the recipient belonged to one of London's largest guilds than when the recipient belonged to a craft guild.

Question 19

With the ongoing expansion of e-commerce, consumers are expecting faster and faster delivery of goods, but delivery companies continue to struggle with last-mile logistics (the final step in delivery to consumers) due to challenges such as complex and inefficient delivery routes. Innovations to mitigate these challenges have been emerging—the use of aerial drones, for example—but these innovations tend to engender their own complications (e.g., regulations on the use of drones in residential airspace), leading researchers to conclude that _____

Which choice most logically completes the text?

- A. consumers' expectations for reduced delivery times may be outstripping what is viable for delivery companies to provide.
- B. a better understanding of consumers' expectations for delivery is needed so that companies can better plan for fluctuations in delivery volume.
- C. rapid delivery is a leading factor in consumer satisfaction, and therefore delivery companies would benefit from investing resources in reducing delivery times.
- D. there may not be sufficient incentive for delivery companies to attempt to solve the problems associated with last-mile logistics.

Question 20

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Vita Sancti Cadoci*, composed in the 11th century, and *Culhwch and Olwen* from the second half of the 12th century. One of the most significant sources, Geoffrey of Monmouth's *History of the Kings of Britain*, was written in the 1130s; some material from it was later adapted by the Norman poet Wace into the *Roman de Brut* in 1155. But Wace didn't merely adapt *History*; he added to it as well, introducing the famous Round Table at which Arthur's knights assembled, which suggests that _____

Which choice most logically completes the text?

- A. *Roman de Brut* is more historically accurate than *History*, because *Culhwch and Olwen* had not been written when Geoffrey of Monmouth was writing his work.
- B. Geoffrey of Monmouth was unaware of stories of the Round Table when composing his *History*, though historians know that works containing such stories were available to him.
- C. Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in *Culhwch and Olwen* than they are to the accounts in *Roman de Brut*.
- D. the Arthurian legends that the author of *Vita Sancti Cadoci* drew on would not have featured the Round Table.

Question 21

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus _____

Which choice most logically completes the text?

- A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.
- B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.
- C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.
- D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

Question 22

One recognized social norm of gift giving is that the time spent obtaining a gift will be viewed as a reflection of the gift's thoughtfulness. Marketing experts Farnoush Reshadi, Julian Givi, and Gopal Das addressed this view in their studies of norms specifically surrounding the giving of gift cards, noting that while recipients tend to view digital gift cards (which can be purchased online from anywhere and often can be redeemed online as well) as superior to physical gift cards (which sometimes must be purchased in person and may only be redeemable in person) in terms of usage, 94.8 percent of participants surveyed indicated that it is more socially acceptable to give a physical gift card to a recipient. This finding suggests that _____

Which choice most logically completes the text?

- A. gift givers likely overestimate the amount of effort required to use digital gift cards and thus mistakenly assume gift recipients will view them as less desirable than physical gift cards.
- B. physical gift cards are likely preferred by gift recipients because the tangible nature of those cards offers a greater psychological sense of ownership than digital gift cards do.
- C. physical gift cards are likely less desirable to gift recipients than digital gift cards are because of the perception that physical gift cards require unnecessary effort to obtain.
- D. gift givers likely perceive digital gift cards as requiring relatively low effort to obtain and thus wrongly assume gift recipients will appreciate them less than they do physical gift cards.